

DETERMINATION PROBLEMS FOR ORBIT CLOSURES AND MATRIX GROUPS

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ORBITS AND ORBIT-CLOSURES

The Orbit Problem (A Reachability Question)

Let G be a finitely generated subgroup of matrices in $GL_d(\mathbb{Q})$. Given $\mathbf{u}, \mathbf{v} \in \mathbb{Q}^d$, is there a $g \in G$ st $g \cdot \mathbf{u} = \mathbf{v}$?

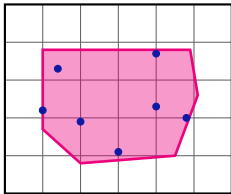
The Orbit Problem is generally *undecidable*, but decidable if G is commutative.

Instead, consider **Orbit closures**...

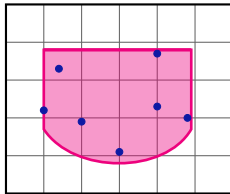
```
 $\mathbf{x} \leftarrow \mathbf{u};$   
while (*) do  
  if (*) then  
     $\mathbf{x} \leftarrow M_1 \mathbf{x};$   
  else  
     $\mathbf{x} \leftarrow M_2 \mathbf{x};$   
  end if  
end while
```

A non-deterministic linear loop

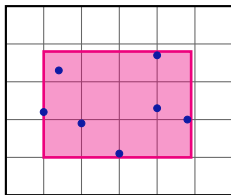
CLASSES OF INVARIANTS



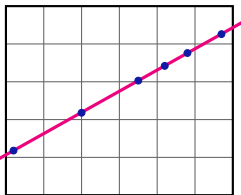
Polyhedrons



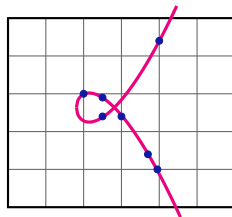
Semialgebraic sets



Intervals



Affine/linear sets

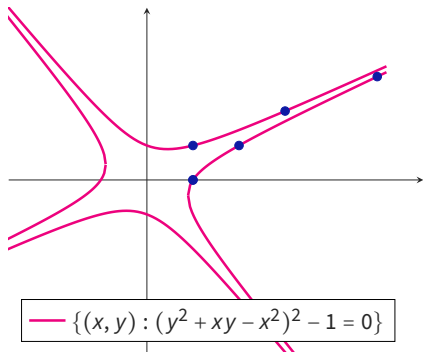


Algebraic sets

An **invariant** is a set that overapproximates the set of **reachable states**.

An **algebraic invariant** is defined by the common zeroes of a family of polynomials.

MOTIVATING EXAMPLE (DYNAMICAL SYSTEM VIEWPOINT)



Input: An algebraic set

$$Z := \{(x, y) : (y^2 + xy - x^2)^2 - 1 = 0\}$$

Find: A matrix-vector tuple^{*}

$$M = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix} \text{ \& } \alpha = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$$

such that $\{\alpha, M\alpha, M^2\alpha, \dots\} \subseteq Z$.

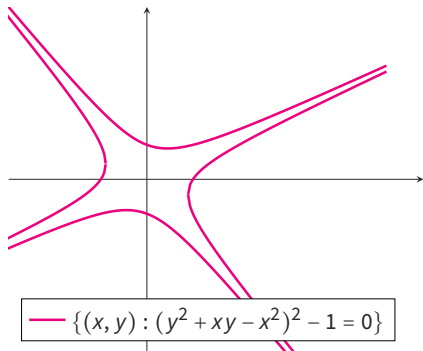
We have $\overline{\{\alpha, M\alpha, M^2\alpha, \dots\}} = Z$

^{*} Find a *maximal* matrix-vector tuple. Otherwise, a valid solution is

$$M = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ \& } \alpha = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

MOTIVATING EXAMPLE (LOOP SYNTHESIS VIEWPOINT)

Called ‘Strong’ Loop Synthesis in [Ait+25] POPL’25.



Input: An algebraic set

$$Z := \{(x, y) : (y^2 + xy - x^2)^2 - 1 = 0\}$$

Find: A deterministic linear loop

$\mathbf{x} \leftarrow \alpha;$

while $(*)$ **do**

$\mathbf{x} \leftarrow M\mathbf{x};$

end while

where Z is the *tightest** algebraic invariant (or Orbit Closure).

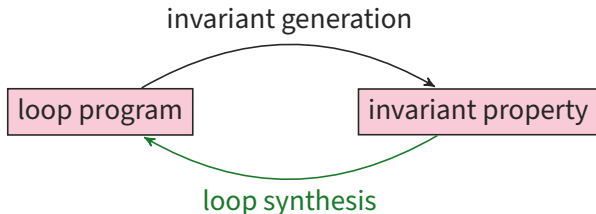
Alternatively: a tuple $\langle M, \alpha \rangle$ with Orbit Closure Z that satisfies the Hoare triples

$\{\text{true}\} \mathbf{x} \leftarrow \alpha \{(y^2 + xy - x^2)^2 - 1 = 0\},$ and

$\{(y^2 + xy - x^2)^2 - 1 = 0\} \mathbf{x} \leftarrow M\mathbf{x} \{(y^2 + xy - x^2)^2 - 1 = 0\}.$

TASK IN LINEAR LOOP SYNTHESIS

Invariant Generation and Loop Synthesis are two sides of the same coin.



“Simple Linear Loops: Algebraic Invariants and Applications” POPL’25

“Linear Loop Synthesis for Quadratic Invariants” STACS’24

“From Polynomial Invariants to Linear Loops” ISSAC’24

“Algebra-Based Reasoning for Loop Synthesis” Form. Asp. Comput.’22

“Algebra-Based Loop Synthesis” IFM’20

THE (CYCLIC) GROUP DETERMINATION PROBLEM

What is known about the analogous (cyclic) group problem?

From the previous slide...

Strong Loop Synthesis or The (Cyclic) Orbit-Closure Det. Problem

Given an algebraic set Z , is there a deterministic linear loop $\langle M, \alpha \rangle$ with $M \in \text{GL}_d(\overline{\mathbb{Q}})$ and $\alpha \in \overline{\mathbb{Q}}^d$ with Orbit Closure Z ?

The (Cyclic) Group Determination Problem

Given an algebraic set $Z \subseteq \text{GL}_d(\overline{\mathbb{Q}})$, is there an $M \in \text{GL}_d(\overline{\mathbb{Q}})$ such that $Z = \overline{\langle M \rangle}$?

Theorem

*There is a procedure that can decide **(Cyclic) Group Determination** and, for a given Z , outputs M in the affirmative case.*

THE OBVIOUS GENERALISATIONS

Loop and group problems

Synthesising Non-deterministic Loops or The Orbit-Closure Det. Problem

Given an algebraic set Z and $s \in \mathbb{N}$, is there a non-deterministic linear loop $\langle M_1, \dots, M_s, \alpha \rangle$ with $M_i \in \text{GL}_d(\overline{\mathbb{Q}})$ and $\alpha \in \overline{\mathbb{Q}}^d$ with Orbit Closure Z ?

The Group Determination Problem

Given an algebraic set $Z \subseteq \text{GL}_d(\overline{\mathbb{Q}})$ and $s \in \mathbb{N}$, is there an s -generated group $G \subseteq \text{GL}_d(\overline{\mathbb{Q}})$ such that $Z = \overline{\langle G \rangle}$?

WE GIVE POLYNOMIAL-SPACE PROCEDURES UNDER THE ASSUMPTION THAT THE MATRICES COMMUTE.

Given an algebraic set Z , can we realize Z as a **matrix group** or as an **orbit closure**, s.t. the underlying group is s -generated?

The Group Determination Problem

Given m polynomials in $\mathbb{Q}[\{x_{i,j}\}_{1 \leq i,j \leq d}]$ of total degree at most b with zero locus $Z \subseteq \mathrm{GL}_d(\overline{\mathbb{Q}})$ and $s \in \mathbb{N}$, is Z an s -generated matrix group?

The Orbit-Closure Determination Problem

Given polynomials $p_1, \dots, p_m \in \mathbb{Q}[\{x_i\}_{1 \leq i \leq d}]$ of total degree at most b with zero locus $Z \subseteq \mathbb{Q}^d$ and $s \in \mathbb{N}$, is Z the orbit closure of some point $\alpha \in \overline{\mathbb{Q}}^d$ under the action of an s -generated matrix group?

```
x ←  $\alpha$ ;  
while (*) do  
  if (*) then  
    x ←  $M_1 \mathbf{x}$ ;  
  else  
    x ←  $M_2 \mathbf{x}$ ;  
  end if  
end while
```

A non-deterministic linear loop

DECIDABILITY OF ORBIT CLOSURE DETERMINATION

Orbit-Closure Determination

Given an algebraic set Z and $s \in \mathbb{N}$, is there a non-deterministic linear loop $\langle M_1, \dots, M_s, \alpha \rangle$ with $M_i \in \text{GL}_d(\overline{\mathbb{Q}})$ and $\alpha \in \overline{\mathbb{Q}}^d$ with Orbit Closure Z ?

Theorem (Polynomial-Space Procedure)

Suppose that $Z = \{\mathbf{x} \in \overline{\mathbb{Q}}^d : p_i(\mathbf{x}), 1 \leq i \leq d\}$ for $p_1, \dots, p_m \in \mathbb{Q}[\mathbf{x}]$ of total degree at most b and $s \in \mathbb{N}$. For Z , **Orbit-Closure Determination** for *commutative matrices* is decidable in time $(mb)^{\text{poly}(d)}$, and space $(d \log b)^{O(1)}$.

```
 $\mathbf{x} \leftarrow \alpha;$   
while (*) do  
  if (*) then  
     $\mathbf{x} \leftarrow M_1 \mathbf{x};$   
  else  
     $\mathbf{x} \leftarrow M_2 \mathbf{x};$   
  end if  
end while
```

A non-deterministic linear loop

For Z , is there a $\alpha \in \overline{\mathbb{Q}}^d$ and s -generated commutative matrix group $\langle M_1, \dots, M_s \rangle$ s.t.
 $Z = \overline{\{M_{i_1}^{n_1} \dots M_{i_k}^{n_k} \alpha : \mathbf{n} \in \mathbb{Z}\}}$?

PROOF FOR ORBIT CLOSURES (COMMUTATIVE CASE)

Orbit-Closure Determination

Given an algebraic set Z , is there a non-deterministic linear loop $\langle M_1, \dots, M_S, \alpha \rangle$ with $M_i \in \text{GL}_d(\overline{\mathbb{Q}})$ and $\alpha \in \overline{\mathbb{Q}}^d$ with Orbit Closure Z ?

1. Under a suitable change of basis, α has entries in $\{0, 1\}$. We guess α in nondet poly time.
2. We want to find subgroups $G_U, G_S \subseteq \text{GL}_d(\overline{\mathbb{Q}})$ s.t. $Z = \overline{G_U \cdot G_S \cdot \alpha}$ (both are s-generated).
3. G_S has a lattice structure Λ in $\mathbb{Z}^k$ (for some k). We guess the generators of Λ in nondet poly time.
4. Existence of G_U reduces to checking the truth of an $\exists^* \forall^* \exists$ -sentence over the theory of real closed fields.

RECAP

The Group Determination Problem

Given m polynomials in $\mathbb{Q}[\{x_{i,j}\}_{1 \leq i,j \leq d}]$ of total degree at most b with zero locus $Z \subseteq \mathrm{GL}_d(\overline{\mathbb{Q}})$ and $s \in \mathbb{N}$, is Z an s -generated matrix group?

The Orbit-Closure Determination Problem

Given polynomials $p_1, \dots, p_m \in \mathbb{Q}[\{x_i\}_{1 \leq i \leq d}]$ of total degree at most b with zero locus $Z \subseteq \mathbb{Q}^d$ and $s \in \mathbb{N}$, is Z the orbit closure of some point $\alpha \in \overline{\mathbb{Q}}^d$ under the action of an s -generated matrix group?

We give polynomial-space procedures for both problems under the assumption that the matrices commute.

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DECISION PROCEDURE FOR FIRST-ORDER SENTENCES

Theorem ([BPR06; CG84])

Consider a first-order sentence in the language of rings that mentions m polynomials in d variables, with total degree at most b , and with k quantifier alternations. The truth of such a sentence over $\overline{\mathbb{Q}}$ and $\overline{\mathbb{Q}} \cap \mathbb{R}$ can both be decided in time $(mb)^{d^{2k+2}}$.

Following [BPR06, Remark 13.10], the truth of first-order sentences, over both $\overline{\mathbb{Q}}$ and $\overline{\mathbb{Q}} \cap \mathbb{R}$, with a fixed number of alternations can be decided in space $(d \log b)^{O(1)}$.

DECIDABILITY OF (WEAK) LOOP SYNTHESIS

Theorem (Ait El Manssour, K, Shirmohammadi, Varonka (POPL'25))

The Loop Synthesis problem over $\{\mathbb{Q}, \mathbb{Z}\}$ is as hard as Hilbert's Tenth Problem over $\{\mathbb{Q}, \mathbb{Z}\}$, even in fixed dimension.

Recall HTP is undecidable over $\mathbb{Z}$, whilst decidability of HTP over $\mathbb{Q}$ is open.